# SALES ANALYSIS DASHBOARD

## 1) Introduction:

The primary objective of the project, which consists of sales order records from an organization, is to provide comprehensive insights into various aspects of the company's sales performance and operations. This data is crucial for informed decision-making and strategic planning.

The dataset includes detailed information about sales transactions, including order dates, customer names, product details, sales amounts, profit margins, shipping modes, and regional data.

**Objective**

- To analyse sales performance across categories, regions, states, and time periods, and provide actionable insights into sales, profit, discount, and order behaviour using an interactive dashboard.
- This dashboard helps management: Monitor sales and profit trends
- Identify top and bottom performing states
- Analyse category-wise sales contribution
- Understand factors influencing sales through key influencers
- Track regional and segment performance

## 2) Methodology

**Data Collection-**Collected sales data containing: Order details, Product category, Region/state, Sales amount, Profit, Quantity, Discount, Customer segment, Order date

**Data Source:** Excel

**Data Preparation**

Performed data cleaning in Power Query:

Removed null values, Corrected data types, Renamed columns, Removed duplicates, etc

**DAX**

Total Sales = SUM(Sales [Sales])

Average Profit = AVERAGE(Sales [Profit])

Total Orders = COUNT (Sales[Order ID])

**Dashboard Development**

**Designed dashboard visuals: KPI Cards**

Total Orders

Average Sales

Average Profit

Average Quantity

Average Discount

Maximum Sales

Minimum Sales

Maximum Profit

Minimum Profit

**Charts Used**

Line Charts → Revenue by Year, Profit by Quarter
Bar Charts → Orders by Category
Scatter Plot → Sales vs Profit by Category
Pie Chart → Sales % by Category
Tables → Top 5 and Bottom 5 states by profit
Key Influencers Visual → Identify sales drivers
Decomposition Tree → Drill-down by Year → Month → Region → Segment

**Tools & Technologies Used**

Power BI Desktop
Power Query
DAX

## 3) Requirement Analysis

### 1) Business Problem

The organization lacked a centralized system to:
track overall sales performance
monitor profit trends
identify high and low performing regions
understand category-wise contribution
analyse factors affecting sales

### 2) Stakeholder Requirements

A) Top Management
They need a high-level summary:
Total Orders
Average Sales
Average Profit
Maximum & Minimum Sales/Profit

**B) Sales Managers**
They need performance tracking & comparison:
Sales trend by Year, Quarter, Month
Revenue growth over time
Regional performance

**C) Category Managers**
They want product/category insights:
Sales by Category
Profit by Category
Order count by Category

**D) Regional Heads**
They require geographical analysis:
Top 5 states by profit
Bottom 5 states by profit
Sales by region (Central, East, West, South)

**E) Analysts / Decision Makers**
They need deep analysis tools:
Key Influencers → what drives sales
Decomposition Tree → root cause analysis

**4) Other Parameters depending upon the Projects**

- Data source identification (Excel)
- Data quality check (missing values, duplicates)
- Data transformation & cleaning
- Business requirements & objectives
- KPI selection based on goals
- Target audience (managers, executives)
- Dashboard performance optimization
- Visualization design & layout

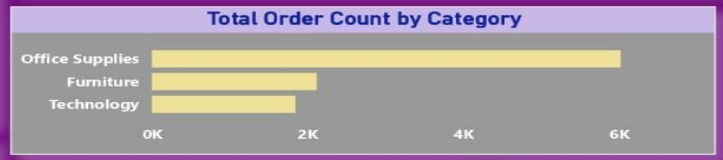
# SALES ANALYSIS DASHBOARD

| TOTAL ORDERS | AVERAGE PROFIT | AVERAGE SALES | AVERAGE QUANTITY | AVERAGE DISCOUNT |
|---|---|---|---|---|
| 9994 | 29 | 230 | 3.79 | 0.1 |

| MAXIMUM SALES | MINIMUM SALES |
|---|---|
| 23K | 0.44 |

| MAXIMUM PROFIT | MINIMUM PROFIT |
|---|---|
| 8.40K | -6.60K |

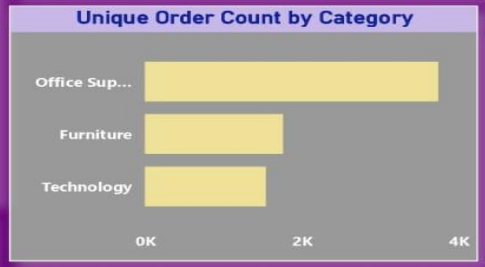
| MIMIMUM DISCOUNT | MAXIMUM QUANTITY |
|---|---|
| 0 | 14 |




Key influencers Top segments

What influences Sales to Increase ?

When... ....the average of Sales increases by

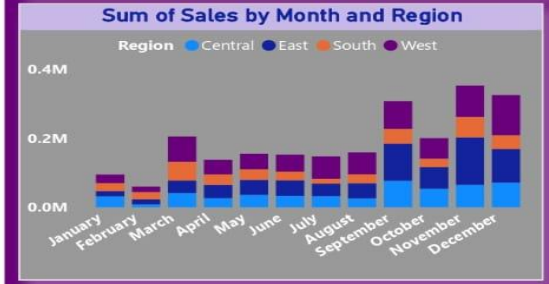


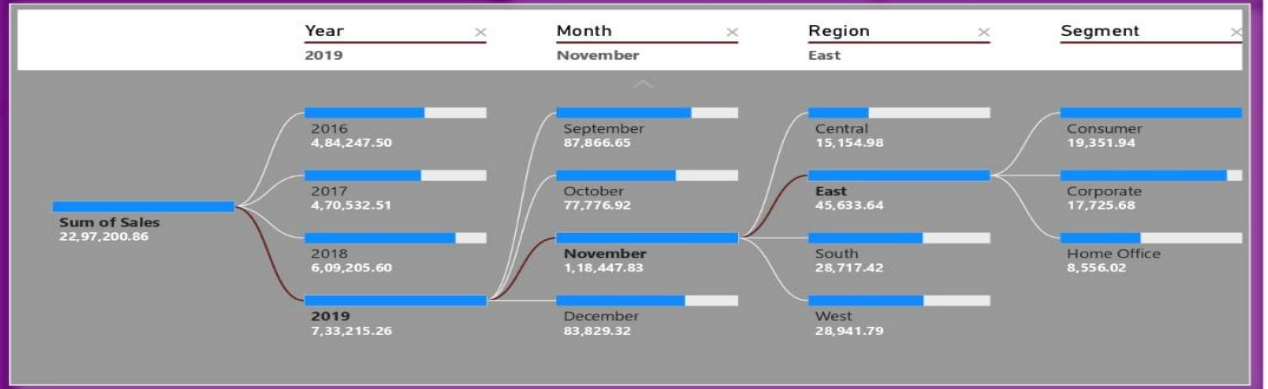

**TOP 5 States by Profit**

| State | Sum of Profit |
|---|---|
| California | 76,381.39 |
| New York | 74,038.55 |
| Washington | 33,402.65 |
| Michigan | 24,463.19 |
| Virginia | 18,597.95 |
| Total | 2,26,883.73 |

**Bottom 5 States by Profit**

| State | Sum of Profit |
|---|---|
| North Carolina | -7,490.91 |
| Illinois | -12,607.89 |
| Pennsylvania | -15,559.96 |
| Ohio | -16,971.38 |
| Texas | -25,729.36 |
| Total | -78,359.49 |



## Insights

**1) Overall Performance Insight**

Total Orders: 9994 → Good transaction volume

Average Sales: 230 → Moderate order value

Average Profit: 29 → Profit margin is relatively low

Business is generating high sales volume but low profit per order, indicating margin pressure.

**2) Sales vs Profit Relationship**

Scatter plot shows:

Some high sales orders have low or negative profit

Higher sales do not always lead to higher profit → discounting or cost issues exist

**3) Profit Trend (Quarter-wise)**

Profit peaked around 2018

Slight fluctuation afterward

Business experienced growth till 2018

**4) Revenue Trend (Year-wise)**

Continuous increase from 2017 → 2019

Strong sales growth trend

But profit is not growing proportionally → margin issue

**5) Regional / State Analysis**

**Top Performing States:**

California

New York

Washington

These are high-profit markets → should be further expanded

**Bottom Performing States:**

Texas

Ohio

Pennsylvania

These states are generating losses → need cost control or strategy change

**6) Root Cause Analysis (Key Influencers / Decomposition Tree)**

Helps identify:

which category, region, or segment drives sales

Business can pinpoint exact factors influencing sales increase or decrease

## Conclusion

**"The dashboard shows that the business is experiencing strong sales growth, especially in the Technology category and key states like California and New York. However, profitability is relatively low due to factors like high discounts and underperforming regions such as Texas and Ohio. While Office Supplies contribute to order volume, Technology drives revenue. The analysis highlights the need to optimize pricing, reduce unnecessary discounts, and improve performance in loss-making regions to enhance overall profitability."**